

Maintenance Schedule

Diesel-electric drilling unit with Series 4000 engine

TPDSZ12V4000-2A0

12 V 4000 G23

Application group 3B

MS50125/01E



Power. Passion. Partnership.

System designation	Engine model	Hz.	kW	1/min	Application group
TPDSZ12V4000-2A0	12V4000G23	50 Hz.	1420 kW	1500 1/min	3B, Continuous operation, variable load, ICXN

Table 1: Applicability

1 Maintenance Schedule

1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance specifications can result in a violation of the statutory emission regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Components relevant to emissions, e.g. control units, sensors, cylinder heads, injectors and exhaust flaps can only be serviced, replaced or repaired if components approved by MTU, or equivalent components, are used.

The MTU maintenance system is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability. The maintenance intervals and required maintenance scopes are based on operational experience and must therefore be regarded as recommendations. In the event of difficult operational and ambient conditions, additional maintenance tasks and/or changes to the maintenance intervals may be necessary.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. The limit which occurs first is applicable.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Component replacement (corrective, and thus not an element of the maintenance schedule).

QL3: Maintenance work that requires partial disassembly of the product.

QL4: Maintenance work that requires complete disassembly of the product.

Supplementary information on maintenance and preservation:

Change intervals for fluids and lubricants and details on preservation can be obtained in the relevant specifications of the component manufacturers and MTU Fluids and Lubricants Specifications. The latest version is available at: <http://www.mtu-online.com>.

Manufacturer's specifications shall apply to the maintenance of all components not listed in this maintenance schedule.

This maintenance schedule does not apply to engines with an EPA/CARB emission certificate.

Application Group

3B Continuous operation, variable load, ICXN

Load factor

Load factor 40%

1.2 Maintenance schedule matrix

0-14,500 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
Engine																															
W1024	1 w																														
W0500	-	X																													
W0501	-	X																													
W0502	-	X																													
W0503	-	X																													
W0505	-	X																													
W0506	-	X																													
W0507	-	X																													
W0508	-	X																													
W0525	-	X																													
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1008	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1247	3 m			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a					X				X				X				X				X				X				X	
W1724	3 a					X				X				X				X				X				X				X	
W1207	18 a			X				X						X						X						X		X			
W1011	4 a													X												X					
W1250	5 a													X												X					
W1006	18 a													X												X					
W1046	9 a																									X					
W1041	9 a																									X					
W2000	9 a																									X					
W2001	9 a																									X					
W2002	9 a																									X					
W2003	9 a																									X					
W2004	9 a																									X					
W2006	9 a																									X					
W2007	9 a																									X					
W2009	9 a																									X					
W2010	9 a																									X					
W2018	9 a																									X					
W2062	9 a																									X					
W2070	9 a																									X					
W2110	9 a																									X					
W2155	9 a																									X					
W1063	18 a																									X					
W1058	18 a																									X					
W1251	5 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
w = weeks m = months a = years																															

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																																					
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500								
W3003	18 a																																						
W3004	18 a																																						
W3005	18 a																																						
W3010	18 a																																						
W3018	18 a																																						
W3042	18 a																																						
W3063	18 a																																						
W3073	18 a																																						
W3083	18 a																																						
W3103	18 a																																						
W3108	18 a																																						
W1544	18 a																																						
Generator																																							
W1258	1 w																																						
W1557	-	X																																					
W1445	-	X																																					
W1406	6 m					X				X				X				X				X				X				X									
W1405	6 m					X				X				X				X				X				X				X									
W1558	6 m					X				X				X				X				X				X				X									
W1409	6 m					X				X				X				X				X				X				X									
W1221	3 a																	X																					
W1559	3 a																	X																					
W1233	3 a																	X																					
W1560	3 a																	X																					
W1561	3 a																																						
W1780	18 a																																						
Fuel prefilter																																							
W1164	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X								
Radiator																																							
W1564	1 w																																						
W1563	-	X																																					
W1565	-			X		X		X		X		X		X		X		X		X		X		X		X		X		X									
W1569	18 a																																						
Preheater / Jacket water heater																																							
W1450	18 a																										X												
W1451	18 a																													X									
W1452	18 a																									X													
W1477	18 a																													X									
Control system																																							
W1571	-			X		X		X		X		X		X		X		X		X		X		X		X		X		X									
Resilient coupling																																							
W1573	-													X																			X						
w = weeks m = months a = years																																							

15,000-24,000 Operating hours

Task	Limit	Operating hours [h]																												
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000										
Engine																														
W1024	1 w																													
W0500	-																													
W0501	-																													
W0502	-																													
W0503	-																													
W0505	-																													
W0506	-																													
W0507	-																													
W0508	-																													
W0525	-																													
W1009	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X									
W1008	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X									
W1247	3 m	X		X		X		X		X		X		X		X		X		X										
W1001	2 a	X		X		X		X		X		X		X		X		X		X										
W1241	2 a			X				X				X				X				X										
W1724	3 a			X				X				X				X				X										
W1207	18 a	X						X						X						X										
W1011	4 a							X												X										
W1250	5 a							X												X										
W1006	18 a							X												X										
W1046	9 a																			X										
W1041	9 a																			X										
W2000	9 a																			X										
W2001	9 a																			X										
W2002	9 a																			X										
W2003	9 a																			X										
W2004	9 a																			X										
W2006	9 a																			X										
W2007	9 a																			X										
W2009	9 a																			X										
W2010	9 a																			X										
W2018	9 a																			X										
W2062	9 a																			X										
W2070	9 a																			X										
W2110	9 a																			X										
W2155	9 a																			X										
W1063	18 a																			X										
W1058	18 a																			X										
W1251	5 a																			X										
W3000	18 a																			X										
W3001	18 a																			X										
W3002	18 a																			X										
W3003	18 a																			X										
W3004	18 a																			X										
w = weeks m = months a = years																														

Task	Limit	Operating hours [h]																							
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000					
W3005	18 a																			X					
W3010	18 a																			X					
W3018	18 a																			X					
W3042	18 a																			X					
W3063	18 a																			X					
W3073	18 a																			X					
W3083	18 a																			X					
W3103	18 a																			X					
W3108	18 a																			X					
W1544	18 a																			X					
Generator																									
W1258	1 w																								
W1557	-																								
W1445	-																								
W1406	6 m			X				X				X				X				X					
W1405	6 m			X				X				X				X				X					
W1558	6 m			X				X				X				X				X					
W1409	6 m			X				X				X				X				X					
W1221	3 a			X																X					
W1559	3 a			X																X					
W1233	3 a			X																X					
W1560	3 a			X																X					
W1561	3 a																			X					
W1780	18 a																			X					
Fuel prefilter																									
W1164	1 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
Radiator																									
W1564	1 w																								
W1563	-																								
W1565	-	X		X		X		X		X		X		X		X		X		X					
W1569	18 a																			X					
Preheater / Jacket water heater																									
W1450	18 a																			X					
W1451	18 a																			X					
W1452	18 a																			X					
W1477	18 a																			X					
Control system																									
W1571	-	X		X		X		X		X		X		X		X		X		X					
Resilient coupling																									
W1573	-							X												X					
w = weeks m = months a = years																									

1.3 Maintenance tasks

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL 1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).		W1024
QL 1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect intercooler drain system.	X	W0502
				Inspect service indicator of air filter.		W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.	X	W0507
				Check differential pressure gauge of fuel prefilter.	X	W0508
				Inspect battery-charging generator for contamination and clean if necessary.		W0525
QL 1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.	X	W1009
QL 1	500	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL 1	1000	3 m	Emergency air shut-off flaps	Lubricate emergency air shut-off flaps at lubrication points.		W1247
QL 1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL 1	2000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.		W1241
QL 1	2000	3 a	Air filters	Check contamination indicator, replace air filter if necessary.		W1724
QL 1	3000	18 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL 1	6000	4 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL 1	6000	5 a	Rubber sleeves	Replace all rubber sleeves.		W1250
QL 1	6000	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL 1	12000	9 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL 3	12000	9 a	Turbochargers	Fit new turbochargers.		W1041
QL 3	12000	9 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.		W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and cam-shaft running surfaces.		W2001
				Clean air ducting.		W2002
				Clean intercooler and inspect for leakage.		W2003
				Fit new high-pressure fuel sensor.		W2004
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.	X	W2009
				Overhaul starter.		W2010
				Clean engine oil cooler and inspect it for leaks.		W2018
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
				Disassemble, clean and check emergency air shut-off flaps (electrical).		W2155
QL 3	12000	18 a	Cylinder heads	Overhaul cylinder heads. Visually inspect piston crowns and wear pattern on cylinder liner running surfaces.		W1063
w = weeks m = months a = years						

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Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	12000	18 a	HP fuel pump	Fit new high-pressure fuel pump.		W1058
QL1	24000	5 a	Hose lines	Replace all hose lines.		W1251
QL4	24000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new fuel delivery pump.		W3010
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Overhaul battery-charging generator.		W3042
				Check gear train for cracks, replace bushings, check axles and replace if necessary.		W3063
				Fit new engine oil pump.		W3073
				Replace wiring harnesses.		W3083
				Replace swing followers and swing-follower shafts.		W3103
				Fit new vibration damper.		W3108
QL4	24000	18 a	Elastomer coupling (main PTO)	Fit new elastomer components .		W1544
Generator						
QL1	-	1 w	Generator: Dust filters	Dust filter: Check for contamination.		W1258
QL1	Daily	-	Generator: Bearing housing	Visually check bearing housing for any sign of grease/oil seepage.		W1557
QL1	Daily	-	Generator	Carry out visual inspection of generator for general condition.		W1445
QL1	2000	6 m	Generator: Electric components	Clean terminal box, cable connection, protective conductor and ground terminals and contact point and check for secure seating.		W1406
				Check insulation resistance.		W1405
QL1	2000	6 m	Generator: Outlet box	Clean the inside of the outlet box, air screens, and air baffles with compressed air or electrical solvent if needed.		W1558
QL1	2000	6 m	Generator: Vibration check	Check for abnormal running noise and vibrations, drain condensate.		W1409
QL3	8000	3 a	Generator: Windings insulation	Check insulation resistance of stator winding.		W1221
QL3	8000	3 a	Generator: Rotating rectifier	Check the rotating rectifier connection tightness.		W1559
QL3	8000	3 a	Generator: Space heaters	Check that anti-condensation heating is working correctly.		W1233
QL3	8000	3 a	Generator: Fan and fan hub	Inspect the fan and fan hub for damage.		W1560
QL3	24000	3 a	Generator: End windings	Remove the endbrackets, and visually inspect the generator end windings for oil or dirt contamination.		W1561
QL4	24000	18 a	Generator	Clean generator and replace ball bearing.		W1780
Fuel prefilter						
QL1	500	1 a	Fuel prefilter	Replace filter element of fuel prefilter and sealing ring depending on the degree of contamination, when the limit (years) is reached, at the latest.		W1164
Radiator						
QL1	-	1 w	Pillow block bearings fan drive system	Regrease pillow block bearings. Do not mix grease types.		W1564
QL1	Daily	-	Fan belt	Check fan belt temperature to ensure that it is within the normal prescribed operating range (If fitted).	X	W1563
QL1	1000	-	Fan guard	Inspect fan guard. Tighten connections as necessary.		W1565
QL4	24000	18 a	Coolant system cleaning	Clean, flush and refill cooling system.		W1569
Preheater / Jacket water heater						
QL1	12000	18 a	Preheater / Jacket water heater	Check and tighten electrical connections.		W1450

w = weeks
m = months
a = years

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
				Check and tighten electrical connections at element enclosure.		W1451
				Check and clean heating element and tank.		W1452
QL 1	12000	18 a	Preheater / Jacket water heater	Overhaul preheater / jacket water heater.		W1477
Control system						
QL 1	1000	-	LOP Dampers	Check the mounting of the LOP dampers. Tighten as needed.		W1571
Resilient coupling						
QL 1	6000	-	Rubber element	Visually inspect rubber element for damage.		W1573
w = weeks m = months a = years						